

TimeJEPA: What 1.1M Parameters and Joint Embeddings Can Do

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Abstract

TimeJEPA is a 1.14M-parameter univariate forecaster pretrained with a joint-embedding predictive architecture (JEPA) and evaluated zero-shot on all 97 GIFT-Eval configurations with one checkpoint, one configuration, and no adaptation of any kind. It reaches 0.863 / 0.596 (MASE/CRPS geometric-mean ratios vs seasonal naive, with sign-flip test-time augmentation; 0.895 / 0.624 without), between Moirai-large (311M) and Moirai-base (91M) on the CRPS ranking, trained on three RTX 3090s. The project’s founding bet was that latent extrapolation is the right pretraining objective for long-horizon forecasting. We pre-registered predictions, ran the paired ablations, and the bet lost: the latent objective is indistinguishable from reconstruction at equal budget, and its apparent long-horizon advantage lives at rollout boundaries, not within the window. Two things won instead. The corpus: every accuracy jump traces to data composition, not to the objective. And the latent as a *judge*: the frozen pretrained energy ranks true futures far above chance, improves TTM-R3 on five of six datasets as a zero-training reranker, and a 4-dimensional residual-statistics latent, gated into the quantile head, carries the entire calibration of the forecast distribution (switching it off costs 18.7 CRPS points; the median is invariant by construction). The corpus makes the forecast; the latent makes the judge. All numbers trace to a dated, pre-registered experimental log.

1 Introduction

This paper reports a model and the fate of the idea behind it. Both matter.

The idea: in I-JEPA [4] the predictor is scaffolding, discarded after pretraining. TimeJEPA inverts that. The predictor extrapolates the future in latent space and is kept as the inference organ; the decoder is a reading head. The bet, registered before any measurement: latent extrapolation is the right pretraining task for a forecaster. Forecasting is generation, not infilling, the causal/masked split of language modeling [15, 34]; the irreducibly unpredictable share of a series’ future is enormous, and a latent loss lets the EMA target encoder absorb it [28]; transfer should be clean.

We tested the bet the way we tested everything: paired runs, one variable each, numeric predictions written into a dated log before results. The bet lost. At equal corpus and budget, latent extrapolation and reconstruction differ by 1.4%, and reconstruction wins most paired cells (Section 4). A monotone long-horizon advantage did appear on GIFT-Eval (+0.8/+3.1/+6.6% over short/medium/long), then a per-step analysis killed the obvious reading: within a window the advantage *shrinks* with depth (slope -1.64% per 100 steps, CI excluding zero). What survives is narrower and unconfirmed: the latent objective may degrade less under iteration on its own outputs.

What moved the numbers instead was data. The full trajectory from 0.979 / 0.677 to 0.863 / 0.596 (Figure 2) decomposes into protocol, geometry, corpus, normalization, synthetic data, batch rationing, checkpoint selection, and one test-time symmetry. The pretraining objective contributed nothing measurable. The field is currently excited about JEPA for time series; our evidence says that as a way to make a small model forecast better, it is not where the leverage is.

The latent earns its keep elsewhere. A pretrained JEPA is implicitly an energy over (context, future) pairs, and that energy is a competent *judge*: it ranks true futures far above chance, improves TTM-R3 on five of six datasets as a frozen zero-training reranker, and a 4-dimensional latent trained to predict residual statistics carries the entire calibration of the quantile fan, an 18.7-point CRPS ablation with the median invariant by construction (Section 8). One sentence summarizes twenty experiments: *the corpus makes the forecast; the latent makes the judge*.

Three evaluation properties frame every number. One checkpoint and one configuration serve all 97 GIFT-Eval configurations; no per-regime variants (the TTM family publishes several). Nothing is tuned, calibrated or selected on GIFT-Eval data. The compute budget is three RTX 3090s; every experiment here is re-runnable by one person on one machine.

2 Related work

Time series foundation models. No paradigm has swept the field: decoder-only (TimesFM [14], Toto [12]), masked encoder (Moirai [43], whose LOTSA corpus we pretrain on), tokenized seq-to-seq (Chronos [3]), xLSTM (TiRex [6]), state space (FlowState [18]), MLP mixers (TTM [16]) all compete on GIFT-Eval [1]. Size is not the lever: Toto spans 4M to 1B parameters, a factor of 250, for 4.6 CRPS points, and sub-10M models (TTM-R3, FlowState, TinyCast) beat much larger ones. That is TimeJEPA’s class, and why testing *which pretraining objective* serves forecasting at fixed small capacity has a place. Other reference points: Lag-Llama [36], Timer [30], Sundial [31], VisionTS [10], YingLong [41], DeepAR [38], TiDE [13].

JEPA and self-supervised learning. JEPA was proposed as a path to predictive world models [28] and instantiated for images and video [4, 5, 9], always with a disposable predictor. Anti-collapse machinery: BYOL’s EMA target [19] with the MoCo v3 momentum schedule [11], VICReg [8], and LeJEPA’s SIGReg [7], our default. Our energy reading of the latent follows LeCun et al. [29]. To our knowledge TimeJEPA is the first systematic attempt to keep the JEPA predictor as the forecasting organ for time series and measure whether that helps.

Components, data, inference. Patching and channel independence follow PatchTST [33]; instance normalization follows RevIN [26], composed with a robust arcsinh transform whose precedent is Toto’s scaler [12]; pre-norm blocks [46] with rotary embeddings [39]. Quantile decoding optimizes pinball loss [17, 27]; metrics follow Hyndman and Koehler [23] and GluonTS [2]. Synthetic pretraining data follows KernelSynth [3] via random Fourier features [35]; CauKer adds causal structure [45]; Toto-2 reports a majority-synthetic recipe [25]; our shape augmentations follow the TiRex ablation [6]. Sign-flip test-time augmentation has precedents in TimesFM and YingLong [14, 41]; conformal calibration follows Romano et al. [37], Vovk et al. [40]; weight averaging [24, 44] failed cleanly for us. The world-model perspective connects to [20, 21, 42, 47]; our measured constraint, prudence from predicted uncertainty rather than from the energy, converges with Henaff et al. [22].

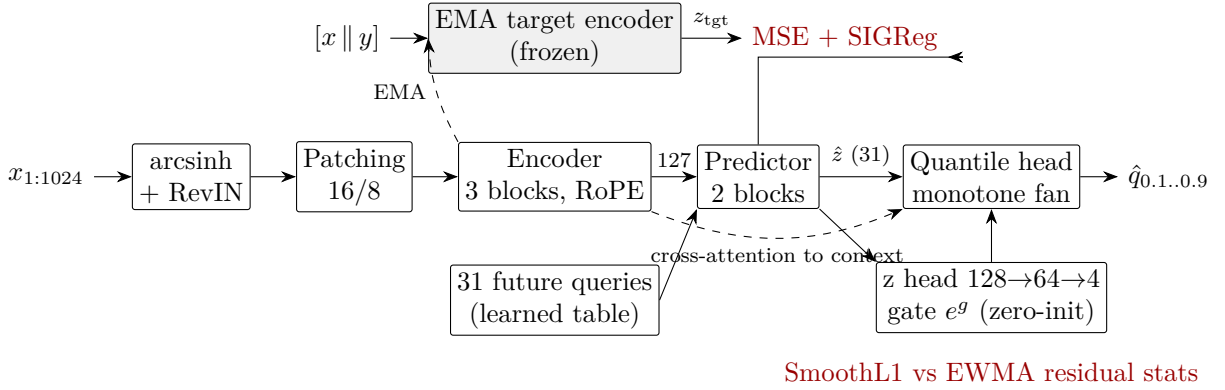


Figure 1: TimeJEPA. Solid path: inference (context 1024 $\rightarrow$ 127 patches $\rightarrow$ predictor with 31 learned future queries $\rightarrow$ nine monotone quantiles for 256 steps). Gray: pretraining target branch (EMA copy, frozen). Red: losses. The z head and its zero-initialized gate are the ESJEPA arm (Section 8.3).

Component	Params	Share
Patch embedding	2,432	0.2%
Online encoder (3 blocks, $d=128$, 4 heads, ff 512)	595,072	52.4%
Predictor (2 blocks, heads, 50-row query table)	419,968	37.0%
Quantile head (cross-attn, MLP, unpatching, gate)	118,162	10.4%
RevIN affine	2	
Trainable total	1,135,636	
EMA target encoder (frozen copy)	595,072	

Table 1: Measured parameter decomposition. The h512 variant adds 6,144 query-table parameters and nothing else.

3 Architecture

Figure 1 shows the full model; Table 1 its parameter budget. No choice below came from taste; each traces to a logged measurement, cited inline.

Geometry, set by measurement. Context 1024, native horizon 256, patch 16 stride 8. A dedicated round took the seven-dataset average MASE from 1.365 to 1.159 and located a flat context optimum (512 to 1280 within 5%, against 33% before). The prior brittleness had a mechanism: skill peaked exactly at the training context length (electricity: +28.5% at 384, −103.8% at 768); the model memorizes a patch count. The cure is per-batch geometry randomization (context left-truncated to $\{128..1024\}$, horizon to $\{64..256\}$, $p=0.5$ each); per-batch keeps tensors rectangular, and RoPE makes the encoder length-agnostic. A scratch control showed the robustness comes from randomization at fine-tuning time, not from pretraining.

Normalization. Two composed stages, fitted on the context only:

$$x' = \operatorname{arcsinh}\left(\frac{x - \operatorname{med}(c)}{s(c)}\right), \quad s(c) = \begin{cases} 1.4826 \operatorname{MAD}(c) & \text{if } > 0.01 \sigma(c), \\ \max(0.1 \sigma(c), 10^{-3}) & \text{otherwise,} \end{cases} \quad (1)$$

then RevIN in compressed space. Each term answers a pathology: MAD/arcsinh for spikes ($\times 100$ spikes in VM traces crush σ -scaled signals; precedent [12]); the floor for near-constant contexts (RevIN scale ~ 0.003 projects targets to thousands of sigmas; arcsinh makes them ordinary numbers); the conditional fallback because an unconditional max would let a spike

re-inflate the scale, the exact pathology MAD exists to ignore. The floor value matters: 29% of bitbrains contexts have MAD exactly zero, and an earlier 10^{-8} floor overflowed float32 through sinh (CRPS $\sim 10^{10}$). Arcsinh is strictly monotone, so quantiles are equivariant and the median commutes with the inverse. The inverse additionally clamps to $[\min(c) - 10w, \max(c) + 10w]$: a half-trained head once emitted $|z| \approx 15$ in compressed space, and sinh turned it into a CRPS of 1.8×10^7 on one configuration, $\times 1.19$ on the whole aggregate; Chronos bounds its vocabulary at $\pm 15\sigma$ for the same reason [3].

One theorem, used twice: median and MAD are 1-homogeneous, so the composed normalization quotients the entire positive affine group. Scale augmentation is provably inert (pinned by test), and the only surviving group element is the sign, which is why flip is the one test-time symmetry worth having (Section 6).

Encoder and EMA target. Three pre-norm blocks [46], RoPE [39], 4 heads, ff 512, non-causal. The target encoder is an EMA copy on the BYOL/MoCo v3 cosine schedule, $\tau : 0.996 \rightarrow 1$ [11, 19].

Predictor. Two blocks (lighter than the encoder, the I-JEPA convention). Input: 127 context embeddings concatenated with 31 learned future queries; the last 31 outputs are the predicted latents $\hat{z}$. The query table is oversized (50 rows) because slicing past a table’s size silently returns fewer rows: in an early version, missing queries were replaced by context embeddings and trained as predictions, corrupting every horizon above 136 steps with valid shapes throughout (Appendix A). Horizon extension at fine-tuning time grows the table by merging trained rows into a longer one. An optional zero-initialized FiLM conditions the queries on a scalar (built for cross-resolution, reused for actions in Section 10); zero-init makes the unconditioned model exactly the baseline.

Quantile head. Nine quantiles per step, three commitments. (1) Direct quantiles because GIFT ranks on CRPS $\approx$ mean pinball over a grid: optimize the metric, no proxy; and MASE is minimized by the median, which pinball at $\tau=0.5$ estimates exactly. (2) Cross-attention back to the context embeddings, because the MSE-trained predictor converges to $\mathbb{E}[z \mid \text{ctx}]$, a conditional mean (measured: predicted variance 0.6 vs target 0.95); dispersion lives in the context, not in $\hat{z}$. (3) Monotone fan by construction: the median is regressed directly, other levels are median $\pm$ cumulative softplus widths; crossing is impossible rather than penalized. The ESJEPA gate multiplies the widths by e^g from a zero-initialized linear map of the predicted residual statistics: exact identity at init, median untouchable, so MASE is structurally invariant and any CRPS delta is attributable to spread alone.

Rollout beyond 256 steps: propagate the fan, not the median. Median feedback has a measured bias: the median path is smoother than any trajectory, so intervals shrink with horizon (exchange h720 band width 0.267 vs a truth growing like $\sqrt{h}$). We propagate all nine levels, copy k continuing from its own level k (comonotone coupling), marginal fan = per-step sort. The assumption, stated honestly: perfect rank dependence across rolls. Deterministic, no RNG, and it moved the average fan gain at h720 from -4% to -16% WQL with band widths finally increasing in depth.

4 Pretraining, and the fate of the founding hypothesis

Objective. Standard JEPA plus one term:

$$\mathcal{L} = \text{MSE}(\hat{z}, z_{\text{tgt}}) + \lambda \Omega(z_{\text{ctx}}) [+ \lambda_z \text{SmoothL1}(\hat{s}, s)], \quad (2)$$

with $\Omega = \text{SIGReg}$ [7] at $\lambda=1$ (16 random 1-D projections pushed toward Gaussian; VICReg control was indistinguishable, 1.193 vs 1.183 average MASE, but SIGReg constrains the whole distribution where a variance hinge accepts bimodal collapse). The bracketed term is the ES-JEPA arm ($\lambda_z=0.1$, Section 8.3). Two implementation lessons that cost real results: compute variance statistics per patch position, not pooled (pooled 0.9990 vs per-position 0.0000, positional diversity masking true collapse), and never regularize a detached tensor (zero gradient, inflated reported loss). AdamW [32], peak LR 3×10^{-4} , warmup 10% then cosine, bf16, batch 512 per GPU on three 3090s. The LR ceiling and the short budgets are measurements, not habits: a cosine calibrated for 40 epochs never decays in 3 (the run oscillated, 0.345–0.445, and its “best checkpoint” was a noise trough), and the full corpus is exhausted in about one epoch, after which validation rises. Collapse is monitored by per-position std and effective rank, calibrated per corpus (rank at init: 3.7–4.8 on Monash, 43–87 on LOTSA).

4.1 The founding bet, tested to destruction

The claim under test: latent extrapolation beats reconstruction as a pretraining objective, increasingly so at long horizon. Four measurements, in order.

1. The paired duel is null. One-variable reconstruction arm (decode to normalized patches, SmoothL1, no anti-collapse: fixed targets cannot collapse), checkpoints paired at 0.4% val loss. Seven-dataset MASE: latent 1.150, reconstruction 1.166. Reconstruction wins 17/28 paired cells, median gap 0.5% in its favor. GIFT-Eval: 0.979 / 0.6766 vs 1.003 / 0.6764, identical CRPS. Verdict: not “JEPA loses” but “indistinguishable,” which suffices to strike the claim.

2. A horizon signal appeared and replicated. Latent advantage monotone in the GIFT term: +0.8% (short, $n=55$), +3.1% (medium, $n=21$), +6.6% (long, $n=21$); same pattern in the Nixtla cells. For a week this was the thesis.

3. The per-step analysis killed the obvious mechanism. If predicting far is cheaper in latent space, the gap must grow with depth *inside* a window. Paired bootstrap over per-step MAE, eight datasets, 20,000 resamples: the gap *shrinks* with depth, slope -1.64% per 100 steps, 95% CI $[-2.82, -0.56]$, negative in 6/8 datasets. Across rollout counts the picture inverts: gap vs number of rolls, Spearman +0.404, $p=0.019$; latent significantly *worse* at h192 (single window), ahead only at h720 (three rolls). Surviving candidate: *the latent objective degrades less under iteration on its own outputs*. Fragile ($p=0.069$ without one dataset), with a registered three-seed protocol not yet funded. A native h512 fine-tune (grown query table, same pretrain, halving the roll count for the affected horizons) supplied a paired check: on configurations unaffected by a window-induced corpus change (Section 7), the multi-rollout group improved by 1.7% relative to the single-roll group, the sign the boundary hypothesis predicts and the magnitude of a minor lever, consistent with the flat term aggregates.

4. What pretraining itself buys. Against from-scratch at equal fine-tuning: in-domain, nothing (1.193 vs 1.187, strict equality); on never-fine-tuned datasets, -26% MASE on 8/8 (upper bound: the strongest dataset was later found contaminated, Appendix A), tightened to -8.7% in a budget-controlled clean rerun. Pretraining does not improve fit; it improves transfer. And note the selector trap: fine-tuning validation loss preferred the wrong model (0.159 scratch vs 0.181 pretrained) while the pretrained one transferred better.

One argument for JEPA survives untouched, and it is architectural, not empirical: latents allow equivariance by objective. A context at resolution k_1 and a target at k_2 can be tied through a predictor conditioned on $w=k_2/k_1$; reconstruction cannot do this, two resolutions do not share target values. Built (the FiLM path), never funded. It is the cleanest reason to keep the formulation.

5 Corpus and batch composition

Most of the accuracy came from here. Working principle: *the batch is the target, the corpus is the means*; every decision is made against an audit of what the sampler actually serves, never against file sizes.

Sources and decontamination. Real data is LOTSA [43], univariate by channel split. Every subset matching an evaluation benchmark is excluded: 7 Nixtla patterns plus 24 GIFT patterns covering all 28 datasets, checked against the official repository tree and pinned by a non-regression test. Exclusion is by whole dataset; exact-name overrides (never substrings) readmit provably disjoint subsets, each justified against all three suites. Cross-check: Salesforce’s GiftEvalPretrain is byte-identical to LOTSA minus 18 datasets; our patterns are strictly stricter, at a quantified cost of 0.9% of observations. PEMS is retained and declared (same road network as `traffic`, no shared sensor, year or frequency). We are strict because we got burned: our original Monash corpus contained `electricity-hourly` and `traffic-hourly`, whose last 20% of steps *are* the Nixtla benchmark; every early number on those two is in-domain, flagged wherever cited, and only the LOTSA lineage is presented as zero-shot (Appendix A).

Chunking. Fixed-length non-overlapping float32 chunks (object arrays break copy-on-write under forked workers; measured 51 GiB), length adapted per subset, capped 2048 for real data. Chunks above 5% missing are dropped; below, linearly interpolated: rejecting any NaN cost 100% of two metro datasets whose nightly closures are regular, and a higher threshold would fabricate 3 a.m. ridership. Windows slide at stride 8 over an int32 index. A per-family cap stops climate reanalysis (era5, cmip6: 63 of 118 audited files) from becoming half the corpus. Short-series GIFT configurations get a pad-to path in v3: left-pad by repeating the first value, so the real data sits where the target is read; the target is always real. Geometry still binds: a 256-step real target cannot come from a 30-point yearly series, so the yearly block stays under-served. Real 5-minute data is mean-pooled by 2 and 3 into legitimate 10/15-minute series.

Sampler, drift, rationing. Batch share $\propto n_i^{0.5}$ per *file*, oversampling cap $3\times$ (at $6\times$, small files overfit visibly). Per-file weighting cuts both ways: climate shards inflated to 66.5% of the batch until capped; later, sharding became our dosing instrument. The cap’s failure mode deserves a paragraph in any pretraining paper: consumed greedily, a family’s cap exhausts early and the family vanishes. Audited: 16 of 71 families extinct before 1% of the epoch, capped-family share drifting $46.6\% \rightarrow 35.7\%$ between first and last decile, batch shrinking $493 \rightarrow 409$. Late training sees a different distribution; a plausible mechanism for late-run GIFT degradation. Fix: *rationing*, a fractional per-batch quota with carry-over. Same exposure, stationary composition across all ten deciles, and the rationed run produced the CRPS champion. Augmentation cannot substitute: it diversifies present samples, not absent families.

Synthetic generator. Three measured holes: nothing between 5-minute and hourly (10T/15T degraded on four datasets), short-series shapes structurally rejected, and no chunks long enough for cross-resolution pairs. The generator follows KernelSynth [3] via random Fourier features [35]: exact GP sampling is $O(N^3)$, prohibitive at $N=8192$; summing $K=64$ sinusoids drawn from the kernel’s spectral density is $O(NK)$ and indistinguishable by eye. Component bank: seasonal blocks with decaying harmonics, smooth GPs, breaking trends, Student- t noise 15% of the time, level shifts, impulses, 25% multiplicative modulation, and a final random affine so synthetic series are not recognizable by their normalization. Table 2 lists the five families and their targets; two have no public-data substitute at all.

Family	Morphology	Target hole	v3 batch
ops_bursty	Poisson bursts to $300\times$, exact zeros, saturation, quantization	bizitobs 10S (ratio 1.387), no public data exists	19.7%
intermittent	integer counts, neg. binomial sizes, seasonal occurrence	car_parts, sparse monthly (our one CRPS loss to SN)	9.9%
subhourly	periods 24–150	10T/15T frequency gap	7.4%
broadband	periods 4–2048	diversity floor	7.7%
lowfreq	periods 4–52, high trend prob.	short-series shapes	5.7%

Table 2: The five synthetic families. Batch shares from the audited v3 composition (54.0% synthetic total, target 50–55%).

Augmentations. Scale augmentation is provably inert under our normalization (Section 3), so the budget goes to shape: amplitude modulation ($p=0.5$), censoring at a random high quantile, same threshold for context and target since saturation is a property of the world ($p=0.5$), and periodic spike trains ($p=0.05$; a random spike in the target alone is unlearnable noise, a periodic one is signal). Order mirrors physics: modulate, spike, then censor. All three are the largest single contributor in the TiRex ablation [6]. Historical note: for the entire first phase the augmentation config was parsed and never wired to the data module; every early run trained without its declared augmentations (Appendix A).

The v3 corpus. Assembled by symlink, nothing deleted: full capped LOTSA (10.05B observations, 4.9M chunks), Chronos-era extras, the padded short-series block, solar_power, the decimated block, and the synthetic families sharded into 23 seeded files (collision with earlier corpora impossible by seed derivation). Because share follows shard count, dose is set by sharding: 3 unsharded families measured 11.2%; the 23-shard allocation was predicted at $\sim 51\%$ and audited at 54.0% after removing two symlinks from a first pass at 55.2%. The audit is the corpus’s identity card: 106 families, stationary across deciles, capped real tail 24.6%. One imprecision reported rather than hidden: the earlier capped corpus was logged as both “3–4 billion” and “ ~ 1.7 billion” observations; the exact count was never instrumented, and the scaling comparison in Section 7 states the range.

6 Evaluation protocol and fairness

One checkpoint, one configuration. Every GIFT-Eval number comes from a single checkpoint and a single model configuration serving all 97 configurations: same weights, same context, same decoding, horizons beyond 256 by the model’s own rollout. No per-regime variants (the TTM family publishes several). Nothing is tuned, calibrated or selected on GIFT-Eval data; the only instrument that reads GIFT outputs is a read-only coverage diagnostic.

Harness. The official protocol is restated next to its upstream sources, not imported: prediction lengths per frequency and term, up to 20 non-overlapping windows over the last 10% of each series, channels split, the 97 configurations pinned as a literal so upstream drift fails loudly. MASE uses each series’ entire past [23]; CRPS is weighted quantile loss over nine levels, pooled per configuration [2, 17]; missing targets masked, never imputed; aggregates are geometric means of ratios vs seasonal naive. We print two normalizations side by side, against the official seasonal-naive numbers and against seasonal naive through our own code path; the gap measures our convention drift (zero on MASE, $\sim 2.7\%$ on CRPS). Context preparation truncates on the left: right-padding would repeat the last value, fabricating observations exactly where they hurt. A full 97-configuration pass takes about four minutes on one 3090 (measured 3 min 30), which is what makes evaluation-based checkpoint selection affordable.

Variant	MASE	CRPS	Verdict
none (champion, plain)	0.895	0.624	reference
sign flip (Eq. 3)	0.870	0.596	official ($-2.0/ -2.4\%$, $2\times$ cost)
scale $f(kx)/k$	=	=	provable no-op (pinned by test)
translation (shifts 1–7)	0.898	0.627	negative: staleness beats phase
multi-lookback {512,1024}	0.896	0.623	mixed alone, worse with flip (0.606)
context 2048 at inference	1.190	0.876	collapse; long context must be trained

Table 3: Test-time variants on the rationed champion (plain baselines where measured). Under a normalization that quotients the affine group (Section 3), the sign is the only symmetry left.

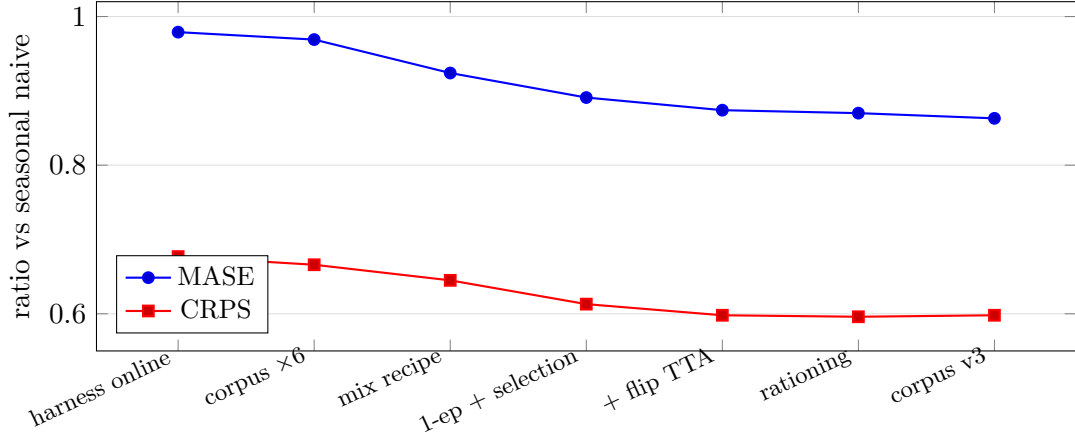


Figure 2: GIFT-Eval aggregates across the project (97 configurations, one checkpoint per point). Every step is a data or inference change; the pretraining objective contributes no step. The last three CRPS points agree to ± 0.002 (Section 7.5).

Test-time augmentation: the sign flip, and nothing else. For a distributional forecast the correct mirror average is

$$\hat{q}_\tau(x) = \frac{1}{2} [q_\tau(x) - q_{1-\tau}(-x)], \quad (3)$$

since $q_\tau(-X) = -q_{1-\tau}(X)$: the fan negates and reverses across levels, ordering preserved. Precedents: TimesFM’s `force_flip_invariance`, YingLong [14, 41]. Table 3 shows why flip is alone: the affine group is exhausted by theorem and by measurement.

Fairness rules. Every aggregate appears plain and with flip, so the reader prices the augmentation separately; single-checkpoint TTA is within the leaderboard’s declared norms and we refuse multi-checkpoint ensembling. No number without its protocol; in-domain lineage labeled as such. Budget: three RTX 3090s, stated wherever a comparison invokes scale.

7 Results

7.1 The progression

Figure 2 and Table 4 give the full GIFT-Eval trajectory with the single change behind each step. Pre-GIFT, on the seven-dataset Nixtla suite: protocol repair set the honest baseline (average MASE 1.365), geometry took it to 1.159, LOTSA to 1.150 zero-shot, the current recipe to ~ 1.110 with wins on 7/7 datasets (skill up to $+41\%$ on traffic).

Date	Step	MASE	CRPS	Change responsible
08-18	first zero-shot (capped LOTSA)	0.979	0.677	harness online
08-21	full corpus	0.969	0.666	corpus $\times 6$ alone
08-22	mix recipe, 15% of epoch	0.924	0.645	synthetic v1 + arcsinh
08-24	mix, 1-epoch cosine, 25%	0.891	0.613	schedule + selection by eval
08-24	+ flip TTA	0.874	0.598	Eq. 3
08-25	rationed run, 45%, flip	0.870	0.596	batch rationing
08-30	v3 corpus, 50%, flip	0.863	0.598	54% synthetic corpus

Table 4: The same trajectory with causes. Plain values of the champions: rationed 0.895 / 0.624; v3 at 50% [pending: plain v3 pair from the evaluation JSON]. Corpus $\times 6$ grows 1.7–4B observations to 10.05B for 1.6% CRPS: volume alone is the weakest lever we measured.

Model	Params (M)	CRPS	MASE	Notes
TinyCast	0.1	0.545	0.774	
TTM-R3	1.4	0.520	0.724	per-regime configs
TimeJEPA (this work)	1.1	0.596	0.863	one config, flip TTA
Lag-Llama	2.4	0.880	1.228	leakage flagged
Toto-2.0-4m	4.1	0.524	0.757	
YingLong-6m	7.3	0.609	0.880	
FlowState	9.1	0.502	0.726	
Kairos-10m	9.9	0.554	0.753	
Moirai2	11.4	0.516	0.728	
Moirai-large	311.0	0.599		adjacent rank
t0-alpha	101.6	0.494		
TimesFM-2.5	231.3	0.490		
Chronos-2	119.5	0.485		
Seasonal naive		1.000	1.000	

Table 5: The sub-12M class (snapshot 2026-08-22; parameter counts and leakage flags are the leaderboard’s own metadata; TTM-R1/R2, Chronos-Bolt and TimesFM-2.0 carry leakage flags and are omitted). TimeJEPA is the only entry trained on a three-GPU consumer budget, among the few with a single configuration for all 97 configurations.

Checkpoint selection by evaluation, not validation. Champions are selected by evaluating checkpoints every 5% of the fine-tuning epoch, because validation loss designates the wrong checkpoint, three times out of three, including once with a perfectly clean, stationary, monotone validation curve (best-val checkpoint 0.896 / 0.631 vs the 45% checkpoint 0.895 / 0.624). Fine-tuning validation and zero-shot transfer measure different things. The selection is honest: fixed checkpoint grid, all 97 configurations, both TTA variants, never a per-configuration choice.

7.2 Leaderboard position

Against the frozen 2026-08-22 snapshot (125 models, official per-configuration CSVs), CRPS 0.596 ranks ~ 91 st, between FLAIR (0.587) and Migas-1.0 (0.598), ahead of Moirai-large (0.599, 311M). Rank trajectory: 108 $\rightarrow$ 103 $\rightarrow$ 99 $\rightarrow$ 92 $\rightarrow$ 91. Table 5 shows the parameter class.

7.3 Where the gap lives

Two floors. The *tail*: sixteen configurations above ratio 1.25 cost ~ 12 aggregate points before the mix recipe; the recipe compressed them to six (~ 7 points), exactly as pre-registered, recovering the predicted configurations (solar 10T from $\times 2.1$ –2.4 to 0.94–0.99; ETT/electricity 15T

from $\times 1.3\text{--}1.45$ to $0.78\text{--}0.98$). The residual tail (bizitobs 10S, electricity/15T/long, us_births) is what the v3 synthetic families target. The *body*: on the other 81 configurations we sit near 0.54 where the sub-10M leaders sit near 0.47, diffuse, no single culprit. Terms are flat (short 0.611, medium 0.619, long 0.616): the rollout is not the bottleneck. Durable strengths: solar weekly beats every model on the board including the 2.5B ones (ratios $0.43\text{--}0.75$); m_dense hourly beats Toto-4m; bitbrains beats TTM and FlowState.

7.4 v3 against its registered bands

Registered before the first checkpoint: plain CRPS success ≤ 0.590 , flip ≤ 0.566 , failure diagnostic at plain > 0.605 . At the 50% checkpoint: MASE 0.8633 flip, best of the project; CRPS 0.5983 flip, 0.24 points off the champion and short of the registered band. P-v3.2 (short-series block) was formally weakened at assembly when geometry proved the point (Section 5). [pending: v3-final: 75–100% checkpoints, verdicts on P-v3.1 (bizitobs < 1.0), P-v3.3 (electricity/15T/long < 1.0), P-v3.4, and the per-configuration autopsy.]

7.5 Triple convergence = a capacity ceiling

Three fine-tunes from three pretrains on three corpora (rationed mix, ESJEPA, v3) land at flip CRPS 0.5959, 0.5981, 0.5983: 0.597 ± 0.002 . Each varied a lever that had previously moved the aggregate; none crossed the floor. The last orthogonal lever, a native h512 fine-tune from the same pretrain (grown query table), confirmed rather than threatened the reading: it plateaus at $0.892 / 0.612$ with flip, its deficit fully accounted for by a mechanical side effect (the 1536-step training window drops the decimated 10-minute block and the padded short-series block from the fine-tuning corpus, costing +11.1% on the 21 affected configurations) plus a horizon effect too small to matter (Section 4.1). It also produced the best-calibrated checkpoint of the project as a side effect, empirical 80% coverage of exactly 0.800, plausibly because fine-tuning under horizon randomization up to 512 steps teaches the head to calibrate across horizons; we flag that ingredient for future recipes. Reading: at 1.14M parameters the model is capacity-bound at this corpus quality, and the next experiment is scale, one variable, with registered success and failure bands. [pending: mini: the 3.4M model against its registered bands.]

8 The latent as judge

The objective is not where forecasting accuracy comes from. Here is where the latent earns its keep. A pretrained JEPA is implicitly an energy function [29]

$$E(x, y) = 1 - \cos(\hat{z}(x), \text{enc}([x \parallel y])), \quad (4)$$

lowered on true pairs by the invariance term. Everything below reads this energy off *frozen* checkpoints: no training on downstream data, anywhere.

8.1 The energy probe

Six GIFT configurations $\times$ 100 instances $\times$ 34 candidates (32 block-bootstraps, seasonal naive, the truth); the metric is the normalized rank of the true future (chance 0.50). Table 6 and Figure 3: pretrained checkpoints are competent judges, and fine-tuning *destroys* the judge.

A second regularity, replicated on three lineages and promoted to a working law: *the judge peaks early*. Best-judge checkpoints sat at mid-run, at 15% and at 5% of their schedules, then eroded while validation kept improving. Forecaster and judge have disjoint temporal optima on the same trajectory; judge selection uses an early cheap probe, never validation and never GIFT.

Checkpoint	Rank of truth ↓	Truth in top 20%	$\rho(E, \text{MAE})$
chance	0.500	0.20	0.00
pretrain, capped corpus	0.245	0.57	0.50–0.74
pretrain, full corpus	0.235	0.60	0.62–0.77
pretrain, mix, mid-run	0.153		(electricity/H median rank 0.000)
pretrain, v3, 5%	0.233		0.507 (standalone record)
<i>full fine-tune</i>	<i>0.409</i>	<i>0.33</i>	<i>erratic, −0.32 to +0.53</i>

Table 6: Energy probe. On strongly seasonal data the pretrained judge is near-perfect (electricity hourly: median rank of the truth 0.030 for the full-corpus checkpoint, 0.000 for mix). The fine-tuned forecaster is a broken judge, below chance on one configuration.

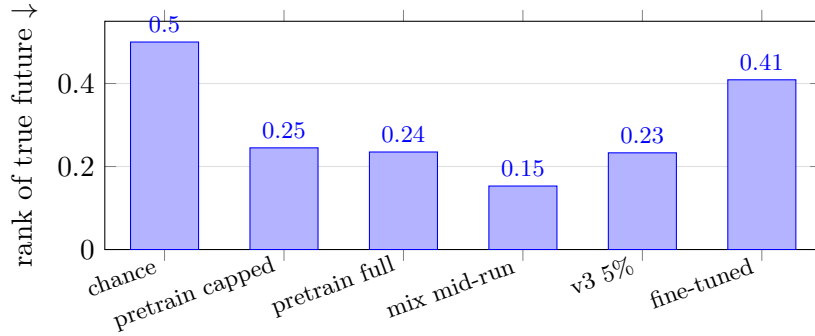


Figure 3: The judge lives in the pretrain and dies in the fine-tune. Lower is better; 0.50 is chance.

8.2 Propose, judge, reweight

The judge becomes useful through a zero-training pipeline: propose candidate futures (block-bootstraps at two block scales, anchors, optionally an external model’s paths), score with E , softmax the *standardized* energies (scale-free, so v0 has no tunable temperature), read weighted quantiles. Table 7 gives the numbers on six Nixtla datasets.

Three mechanisms behind the table. The bootstrap’s failure is its envelope (it recombines the past; exchange trends out of it), which the decoder repairs as proposer. Candidate refinement by a few gradient steps on E (planning by backprop, small step count as a Goodhart guard) adds 1–2% where the envelope binds; on the centered pool below, where no envelope binds, we measured it null on five datasets and slightly harmful on the trending sixth, so refinement and centering are substitutes, not complements, and the official pipeline uses centering alone. And the deliverable is *uplift over the proposer*: a verifier for forecasts, not a competing forecaster. Generation is hard, verification is easy; a frozen judge that improves a state-of-the-art proposer with zero dedicated training is, to our knowledge, a novel use of a JEPA latent in time series.

Killing the dilution was a matter of center, not of weighting. The dilution signature survived a scalar temperature, a calibrated temperature, and three proposer changes; it died the day the candidates changed *center*. A history bootstrap is centered on the past, so every gram of weight it receives pulls the weighted median away from the proposer’s trend. The fix in the bottom block of Table 7 bootstraps seasonal innovations $y_t - y_{t-m}$ and re-attaches the blocks to the proposer’s path: spread without a competing center. Registered predictions held on both counts (the two dilution datasets recover; the four wins keep or grow their margins), and one safety property survives by design: seasonal naive stays in the pool as an alternative center, and in a controlled test with a deliberately biased proposer the judge moves its mass

WQL vs seasonal naive ↓	ettm1	ettm2	etth1	etth2	weather	exchange
energy only (bootstraps + judge)	1.00	0.80	0.76	0.92	0.96	1.15
our decoder (fine-tuned)	0.92	0.74	0.70	0.92	0.76	0.77
decoder proposes, judge weighs	0.95	0.75	0.68	0.91	0.82	0.90
TTM-R3 alone [16]	0.86	0.79	0.85	0.95	0.70	0.88
TTM proposes, our judge weighs	0.84	0.74	0.72	0.88	0.85	0.97
TTM-R3 alone (v3-judge run)	0.88	0.81	0.85	0.95	0.63	0.88
+ judge, proposer-centered pool	0.77	0.72	0.71	0.78	0.64	0.78

Table 7: Propose-judge-reweight, WQL ratios vs seasonal naive (lower is better; bold marks the winner within each block). Middle block: the original pool (raw history bootstraps), where a frozen 1M pretrain improves TTM-R3, the strongest sub-10M model on GIFT-Eval, on 4/6 datasets, with *dilution* on the other two, a signature that replicated four times: when the proposer is already excellent, history-centered bootstrap candidates pull the weighted median toward the past. Bottom block: the fix, registered with numeric predictions before the run. Bootstrapping seasonal innovations and re-attaching the blocks onto the proposer’s own path makes every candidate share the proposer’s center, so center dilution is impossible by construction; the drift anchor leaves the pool, and a degeneracy guard returns uniform weights when energies carry no signal. The hybrid then beats or ties TTM on 6/6 (weather within one point), exchange swinging from a 20-point loss to a 10-point win with its point forecast restored. Judges from two unrelated pretraining lineages replicate the uplift.

there, three standard deviations of preference. A verifier that can fire its proposer is worth having.

The GIFT transplant took two attempts, and both teach. The naive one failed mechanistically: a TTM-paths-only pool collapses because TTM contracts input jitter (near-identical paths, fan degenerating to a point, 80% coverage 0.206), and near-identical energies turn the standardized softmax into a noise amplifier that dumps mass on anchor candidates, with localized disasters exactly on the daily and weekly configurations where a linear drift anchor diverges. With both mechanisms guarded (drift out of the pool, degenerate energies falling back to uniform weights) and the centered pool in place, the pipeline transplants: over all 97 configurations on paired capped windows, the hybrid’s fan beats TTM’s collapsed point forecast by 7.2% CRPS on identical instances at a 3.8% MASE tax (ratio 0.794 against TTM’s 0.765), with coverage improving from 0.206 to 0.622. The remaining honest gap is that coverage: seasonal innovations understate spread at long horizons, where the proposer’s rollout is deterministic past its first segment. The TTM-alone column doubles as harness validation (0.765 on capped windows against the official 0.724), and one side finding: TTM emits NaN on 100% of two sparse count configurations (hierarchical_sales and restaurant, daily) where our own model runs unaided.

8.3 ESJEPa: predict the statistics of what cannot be predicted

The predictor converges to a conditional mean and carries no dispersion (Section 3). The fix is not predicting the residual (irrecoverable) but its *statistics*: a 4-dimensional deterministic target per patch (log-RMS and log-MAD of the residual around a causal EWMA, bounded skewness, lag-1 autocorrelation), predicted by an 8.5k-parameter head on the predictor trunk and injected only through the zero-initialized width gate. Nothing can collapse: the target is a fixed function of the data. Pre-registered witnesses held: predicted-vs-realized log-RMS correlation 0.63 (kill threshold 0.3); prediction-to-target std ratio 0.7 against 0.3–0.4 for the signal path, a finding in itself (*volatility is more predictable than its realization*); the twin run’s invariance loss 4–5% *better* with z , the feared siphoning inverted.

Configuration (flip TTA)	CRPS	80% coverage	MASE
champion, no z (rationed 45%)	0.596	0.769	0.870
ESJEPA 45%	0.598	0.790	0.874
ESJEPA 5% (early)	0.626	0.800 (nominal)	0.921
ESJEPA 5%, gate bias $\rightarrow 0$	0.626	0.813	0.921
ESJEPA 45%, <i>gate off</i>	0.785	0.968	0.873

Table 8: ESJEPA ablations, all by evaluation-time weight surgery (no retraining). The gate-off row is the headline: +18.7 CRPS points and coverage exploding to 0.968 (q10 at 0.014, q90 at 0.982) while MASE moves by $< 10^{-3}$, the median invariance verified in production. The bias-only surgery shows the effect is conditional, not a constant: solar 10T keeps its full -30% CRPS either way.

The decision rule was registered before results: (a) beat the champion, z enters the bundle; (b) within 0.5 CRPS points with materially better coverage, z stays out, becomes the paper ablation; (c) worse, attributed without appeal. Outcome: (b). Table 8 carries the section’s main result.

Read the table bottom-up. The gate does not modulate the fan; it *drives* it: fine-tuning co-adapted the organs, base softplus widths converging to a near-worst-case envelope and the gate to conditional sharpening (negative gains on calm windows, the opposite of the widener we designed). Within this model, four latent dimensions carry the entire calibration. Two honesty notes: the counterfactual trained without z is the champion at 0.596, so z is not worth 18.7 leaderboard points, it is worth nominal coverage at equal CRPS; and “z on a healthy pretrain” is open, not refuted, an ablation the v3 lineage can host.

8.4 What the judge knows

On a fully owned testbed (linear thermostat, hidden action, history under a bang-bang policy: observational confounding in miniature), discrimination AUCs: time-reversed futures **0.808**, morphology violations **0.935**, state-jump discontinuities **0.556**, chance. The judge knows the arrow of time and the shape of trajectories; it accepts a teleported future. Consistently, E as a planning *regularizer* fails a fair test: the dormant term changes nothing (violations 78.7% vs 80.0%), the active term hurts (78.7% $\rightarrow$ 82.4%). Three concordant runs: *the energy is a classifier, not a regularizer*. Planning prudence must come from predicted uncertainty (fan + z), converging with Henaff et al. [22].

9 Clean negative results

Each row of Table 9 was a hypothesis we considered plausible enough to build and measure, with a registered expectation where the log records one. Several contradict folk practice; that is why they are a section, not a footnote.

10 Perspective: one arm, two trades

The judge results reframe the predictor: it is already a world model of an *autonomous* system. The completion, and the original promise of the name, is an action input, unifying forecasting and predictive control in one arm. The design is fixed by measurements already reported.

Mechanism. An action embedding conditions the predictor through zero-initialized FiLM, exactly the pattern of Section 3: $a = \emptyset$ is the identity, so the actioned model *is* the current forecaster, GIFT behavior unchanged. A spectator has no action: marginal dynamics, already

Hypothesis	Measurement	Verdict
Latent objective beats reconstruction	1.4% average, recon wins 17/28 cells; within-window slope $-1.64\%/100$ steps, CI excl. 0	refuted as stated; rollout-boundary candidate open (§4.1)
Weight soup of the selection window [44]	0.878 / 0.607 flip vs 0.596 single; m4_weekly $2.76 \rightarrow 4.13$	negative; the average leaves the basin; closed
Uniform conformal calibration [37] (registered: -1 to -2.5%)	$\gamma \approx [1.01..1.12..1.03]$; CRPS 0.5961 vs 0.5959; MASE bit-identical	no-op; the fan is already calibrated in-distribution, GIFT under-coverage is <i>shift</i> , exactly what the ESJEPa gate addresses
Scale TTA	bit-identical inputs	provable no-op, pinned by test
Translation TTA	0.898 / 0.627 vs 0.895 / 0.624	negative; staleness wins
Context 2048 at inference	1.190 / 0.876; yet bizitobs 5T CRPS halves	fails globally; long context must be trained (seeded the v3 curriculum)
Energy as planning regularizer	dormant: no effect; active: violations $78.7 \rightarrow 82.4\%$	refuted, 3 concordant runs (§8.4)
Corpus volume alone	$\times 6$ data $\rightarrow -1.6\%$ CRPS	weakest lever measured
Validation loss selects checkpoints	wrong 3/3, once with a clean curve	doctrine: select by evaluation
Stricter-than-required decontamination	0.9% of observations	empty lever, quantified

Table 9: Negative results with their numbers. Two learning-rate lessons belong here too: a cosine calibrated for 40 epochs never decays in 3, and a full corpus is exhausted in one epoch (validation rises afterward).

learned. An actor knows its candidate action by definition. The controller is not a network; it is inference-time optimization, planning by backprop (the refinement machinery of Section 8.2 with the action as the variable) or propose-and-judge over plans [21, 42]. The predictor never decides; it simulates, the loop decides.

Three measured constraints. Prudence comes from fan + z, not from E (the judge accepts teleported futures and hurts plans as a cost term, Section 8.4; convergent with Henaff et al. [22]). State continuity must be an explicit cost term, since the energy will not supply it. The judging checkpoint is not the forecasting checkpoint and is selected by an early probe (Section 8.1).

The wall is data. Observational corpora carry no labeled actions and are confounded: a model trained on thermostat logs learns that heating turns on when temperature drops, and a naive planner cuts the heating to warm the room. We measured this in miniature on the owned simulator (plans pulled toward historical setpoints). The exit is what the synthetic pipeline makes cheap: causal generators with labeled interventions [45] grafted onto the v3 machinery. The through-line: an unobserved action *is* a latent variable explaining the future, the z of LeCun [28]; ESJEPa is its deterministic shadow, the action input its declared form.

The claim this aims at. A quantile-fan TSFM whose native output supports chance-constrained MPC, zero-shot, on unseen systems, with $a = \emptyset$ leaving its benchmarks untouched. No current TSFM occupies that position; every component (fan, gate, judge, refinement, generator pipeline) exists in this codebase. Sequence: action FiLM, causal corpus, then the planner that is already written.

11 Limitations

One seed per arm, everywhere, including the objective duel; the registered three-seed protocol is unfunded. We mitigate with paired designs, bootstrap intervals and pre-registered directional predictions; we do not solve it. The body gap is real: 0.54 vs 0.47 on the 81 non-tail configurations, and the 0.597 ± 0.002 triple convergence reads as a capacity bound at 1.14M, an interpretation until a larger model tests it. The harness is a restatement of the official protocol with measured convention drift (zero MASE, $\sim 2.7\%$ CRPS), and all ranks are against one frozen snapshot (2026-08-22). The pre-v3 corpus observation count was never instrumented (1.7–4B). The short-series block remains geometry-bound; the registered fix (crop-and-pad augmentation) is designed, not built. Scope: univariate by choice; no covariates; judge results are measured on this architecture’s checkpoints only.

12 Reproducibility statement

The project runs against a dated experimental log: hypotheses, configurations and numeric success/failure bands are written before each run, results appended after, negative ones included. This paper renders that log; every number traces to an entry and a commit. Configurations are declarative one-variable overlays; the evaluation harness caches per-configuration JSON keyed by a checkpoint fingerprint, so a stale checkpoint cannot silently reuse old results (Appendix A). The corpus assembles by symlink from immutable sources with an audited batch composition; the generator’s seed scheme makes cross-corpus collisions impossible.

Compute, in full: three RTX 3090s. A pretraining epoch is one to two days; fine-tuning a fraction of one epoch; a 97-configuration GIFT pass under four minutes on one GPU; probes and the planning testbed run on CPU. Total, including every failed run: a few GPU-weeks of consumer hardware. This is part of the claim, not an apology: every experiment here is re-runnable by one person. The test suite (295 tests) pins the mechanisms that carried silent failures; as Appendix A shows, tests guarantee mechanisms, not protocol coherence, and the log is the second, equally necessary instrument.

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A Silent failures, and how they were actually caught

Most defects that invalidated results were silent: valid shapes, decreasing losses, no warnings. Three of the worst were caught by reading log lines, not by tests. Table 10 is the catalogue; we consider it as reusable as any architecture choice.

Defect	Consequence	Caught by
Eval-side global normalization vs per-instance training	MSE overstated 42% avg over 40 pairs; nothing pre-repair usable	protocol audit
RevIN affine inverted at decode, never applied forward	6–10% scale error + offset on every forecast	code reading
Query table sliced past its size	missing queries silently replaced by context embeddings, trained as predictions; every horizon > 136 corrupted	code reading; now refused
VICReg pooled over positions; half on a detached tensor	positional diversity masked true collapse; zero-gradient regularizer	measurement (0.9990 vs 0.0000)
Optimizer built before gradual unfreeze	unfrozen weights never stepped; every gradual run was a frozen-encoder probe	log line: parameter counts
Augmentations parsed, never wired	first-phase runs trained without their declared augmentations	log line
<code>last.ckpt</code> = last top- k copy, not last state (Lightning version difference)	mid-run checkpoint served as “final”; the operator had measured it, review dismissed it from another version’s source	18 bit-identical probe values; md5. Measurement beats reading code. Parade: save every checkpoint
Corpus contained the benchmark	electricity/traffic-hourly extend the Nixtla series over their eval period; early transfer claim downgraded	found while writing exclusion rules for another corpus
Cached eval resumed across an overwritten checkpoint	old model’s numbers reported as new, “already done” $\times 97$	fingerprint now in cache key

Table 10: Ten defects, none caught by the test suite that now pins them.

The pattern: correctness of mechanism and coherence of protocol are different properties. Tests grew to pin each mechanism after the fact; what remained was protocol-level and was caught by two near-free habits. Pre-registered numeric expectations: a result that contradicts its prediction triggers a search that a merely good-looking result never would. And reading the run’s own log lines (loaded-key counts, skipped-file lists, per-family batch shares) instead of trusting dashboards.